

Skills Summary

Building Equivalent Ratio Tables

Use this reference while completing your worksheet or when studying.

Scaling a Ratio Table Up or Down

Rule: Every column of a ratio table shows the **same** ratio. To build a new column, multiply or divide **both** quantities by the same number (the multiplier). Never add the same amount to both — adding changes the ratio.

Example: A ratio table starts with 2 stickers for 5 cents. Find the cents column for 6 stickers.

Step 1. Find the multiplier from the known column: $6 \div 2 = 3$, so both quantities are multiplied by 3.

Step 2. Cents: $5 \times 3 = 15$, so 6 stickers cost **15** cents.

Try it: A table starts with 3 goals for 4 saves. Find the saves value in the column for 9 goals.

check: 12

(!) Watch out: The most common table mistake is **adding** the same number to both rows. Check any new column by simplifying: it must reduce to the original ratio.

Building a Unit-Rate Column

Rule: The most useful column in a ratio table is the column for **one** unit. Divide both quantities by the first quantity to build it; then any other column is just the unit column times the amount you want.

Example: 3 pens cost 12 dollars. Build the one-pen column.

Step 1. Divide both quantities by 3 so the pen count becomes one.

Step 2. Cost: $12 \div 3 = 4$, so one pen costs **4** dollars.

Try it: 6 erasers cost 18 dollars. How many dollars does one eraser cost?

check: 3

Finding a Missing Value with a Proportion

Rule: Two equivalent ratios written as fractions make a **proportion**. To find a missing value, find what one part was multiplied (or divided) by, and apply the same multiplier to the other part.

Example: Find the missing value: $\frac{4}{9} = \frac{?}{36}$.

Step 1. The denominator was multiplied by 4, because $9 \times 4 = 36$.

Step 2. Multiply the numerator by the same 4: $4 \times 4 = 16$, so the missing value is **16**.

Try it: Find the missing value: $\frac{5}{6} = \frac{?}{42}$.

check:

Key Vocabulary

Ratio table: a table whose columns all show the same ratio **Unit rate:** the value paired with one unit

Multiplier: the number both quantities are scaled by **Proportion:** an equation stating two ratios are equal